

Multiple Choice (10 marks)

1. Safe practices in the laboratory include all of these except
 - (a) tying back long hair.
 - (b) wearing safety goggles.
 - (c) performing each step quickly.
 - (d) wiping up spills immediately.
2. In a forest, how do decomposers help other organisms survive?
 - (a) They release oxygen into the air that animals breathe.
 - (b) They put nutrients into the soil that plants use to grow.
 - (c) They provide shelter in forests where animals can hide.
 - (d) They use sunlight to make food for plants and animals.
3. The Moon lacks weather and climate changes like those on Earth. What causes the lack of weather on the Moon?
 - (a) the lack of water
 - (b) the presence of igneous rock
 - (c) the very thin atmosphere
 - (d) the lack of magnetic poles
4. Della has a mixture of soil and water in a jar. Which of the following tools would best help Della separate the soil from the water?
 - (a) a filter
 - (b) a measuring cup
 - (c) a balance
 - (d) a magnifying glass
5. Most mountains on Earth occur in huge ranges stretching for thousands of miles. Which event is most likely to cause the formation of these mountain ranges?
 - (a) erosion of rocks
 - (b) violent earthquakes
 - (c) volcanic eruption
 - (d) movement of plates

True / False (10 marks)

Answer True or False

6. True or False: A human sperm cell contains 23 chromosomes, which is half the number found in a typical human body cell.
7. True or False: The speedometer in an automobile provides feedback to the driver about the vehicle's current speed.
8. True or False: Sexual reproduction in daffodils increases the diversity of inherited traits within their population.
9. True or False: An increase in bison population will lead to more tall shrubs and trees in their grazing area.
10. True or False: North America receives less direct sunlight during the winter, resulting in colder temperatures.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of the body's regulatory systems in maintaining a normal body temperature of 98.6° *including the mechanisms involved and their importance for overall health.*
11. Discuss how the physical appearance of a mineral can provide insights into its properties and classification, and explain why visual characteristics are important in mineral identification.

End of Paper